

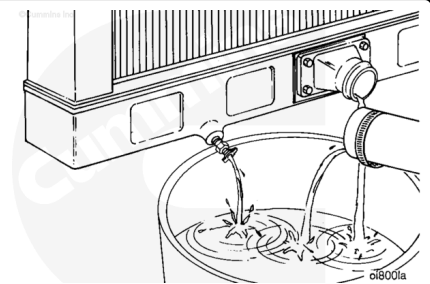
Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

This procedure is designed to remove the two-stage assembly from a QSK60 two-stage engine with a minimal amount of disassembly and requires a minimum of 305-mm [12-in] overhead clearance.

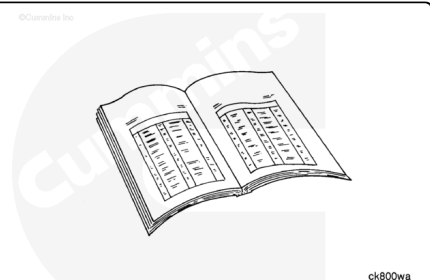
The approximate weight, after removal of suggested components, is 885 kg [1951 lb].

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

with Electronically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).
- Remove the turbocharger oil supply lines. Refer to Procedure 010-046 (</qs3/pubsys2/xml/en/procedures/56/56-010-046-tr.html>).
- Remove the turbocharger oil return lines. Refer to Procedure 010-045 (</qs3/pubsys2/xml/en/procedures/56/56-010-045-tr.html>).
- Remove the turbocharger cooling hoses. Refer to Procedure 010-041 (</qs3/pubsys2/xml/en/procedures/56/56-010-041.html>).
- Remove the intercooler assembly. Refer to Procedure 010-083 (</qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html>).

Remove

with Mechanically Actuated Injector

With Mechanically Actuated Injectors

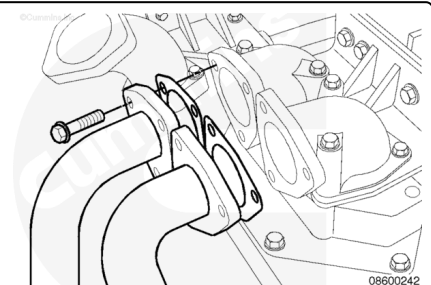
This procedure is designed to remove the two-stage assembly from a QSK60 two-stage engine with a minimal amount of disassembly and requires a minimum of 305-mm [12-in] overhead clearance.

The approximate weight, after removal of suggested components, is 885 kg [1951 lb].

It is a good practice to mark the location of all lines during removal.

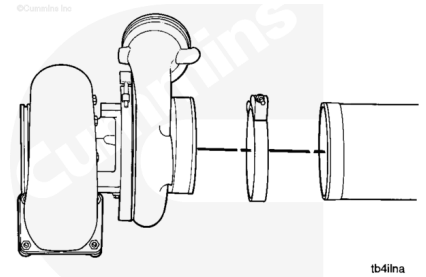
Disconnect the coolant lines from the thermostat housing.

Position the coolant lines aside.



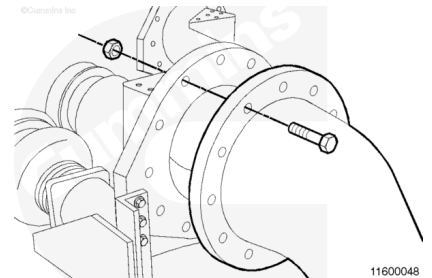
Disconnect the air intake piping between the air filter(s) and low pressure turbochargers.

Disconnect the air intake piping at the inlet side of each turbocharger.



Remove the 12 exhaust outlet flange capscrews and nuts.

Disconnect the exhaust piping from the exhaust outlet flange.

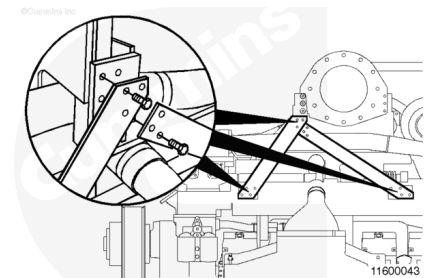


Remove the supports.

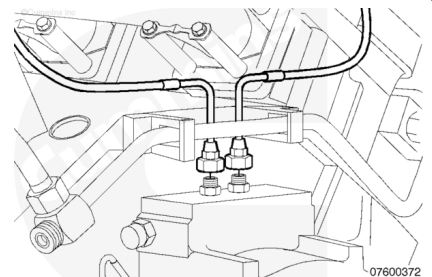
Remove the 12 capscrews connecting the air crossover to the aftercooler cover.

Remove the clamps from the hose connecting the air crossover to the high pressure turbocharger.

Remove the air crossover.

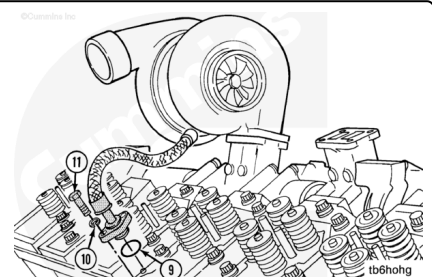


Disconnect the oil supply lines from the flywheel housing.



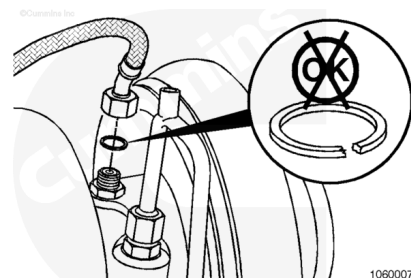
Disconnect the oil drain lines at the cylinder block for all eight turbochargers.

Remove and discard the o-ring seal (9).

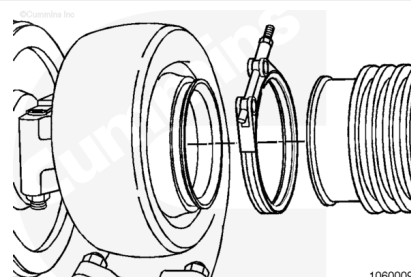


It is a good practice to mark the location of all lines during removal.

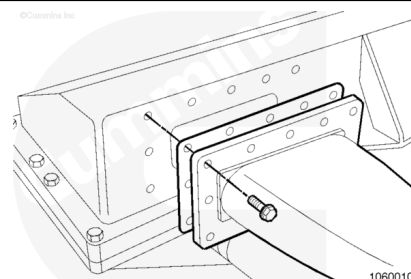
Disconnect the oil supply lines from each of the four high-pressure turbochargers.



Disconnect the V-band clamp at the bellows on the turbine side of the four high-pressure turbochargers.

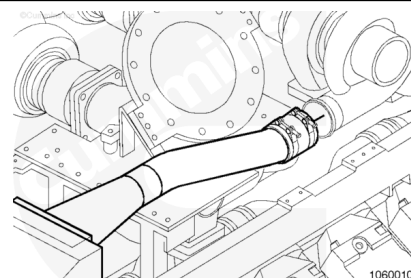


Remove the 12 capscrews from each of the four intercooler intake crossovers to the low-pressure turbochargers.



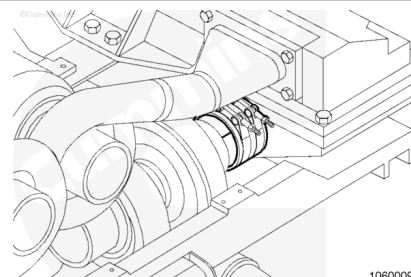
Loosen the hose clamps on each end of the four air connection hoses for the four intercooler intake crossovers and the four low pressure turbochargers.

Note : The left rear low-pressure turbocharger connection is shown. The other low-pressure turbocharger connections are similar.



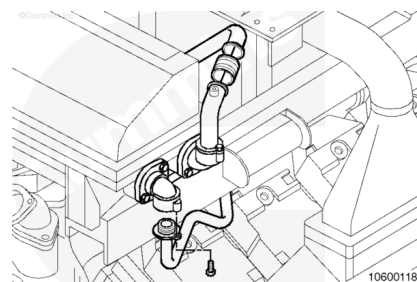
Loosen the two intake hose clamps each on the four high pressure turbochargers and the two intercoolers.

Disconnect the hose clamp at the intercooler.

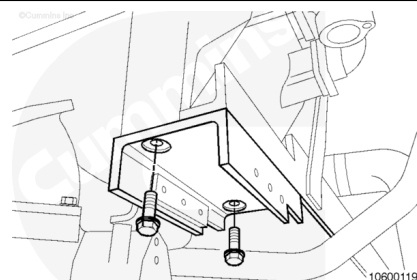


Note : The front intercooler water supply and return connections are shown. The rear intercooler connections are similar.

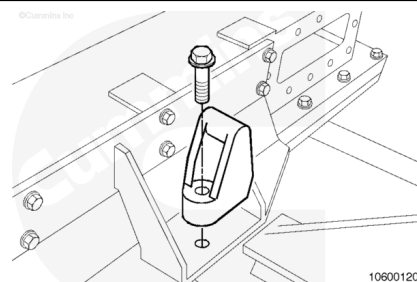
Disconnect the water supply and return piping from the intercoolers.



Remove the capscrews from each of the four feet of the intercooler.



Remove the lifting eye from the bracket between the intercooler and the two-stage frame.

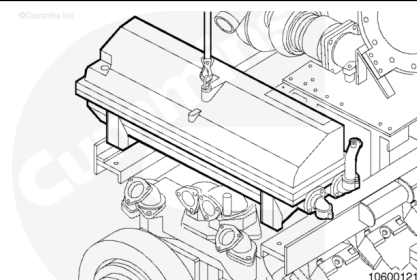


⚠ WARNING ⚠

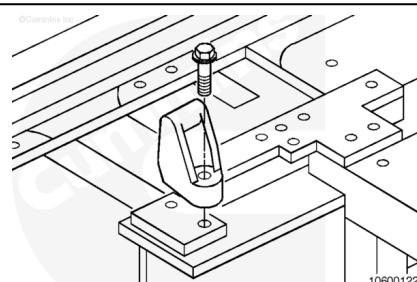
This assembly weighs 23 kg [50 lb] or more. To reduce the possibility injury, use a hoist or get assistance to remove this assembly.

Install a 12 mm [0.47 in] lifting device in the top of the intercoolers.

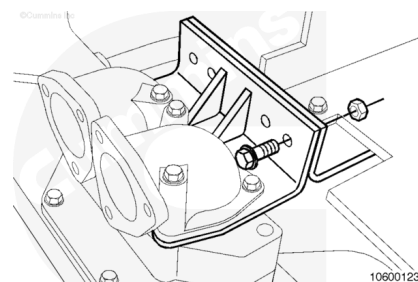
Remove the intercoolers.



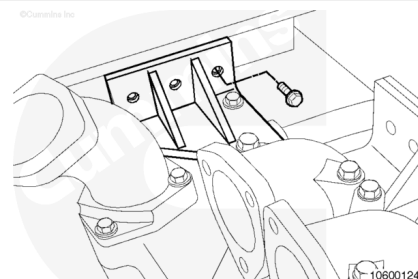
Install the two lifting eyes on the two-stage frame.



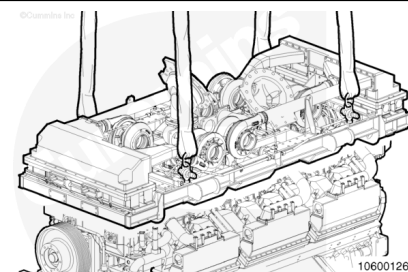
Remove the four capscrews from the bracket connecting the low temperature aftercooling housing to the two-stage frame.



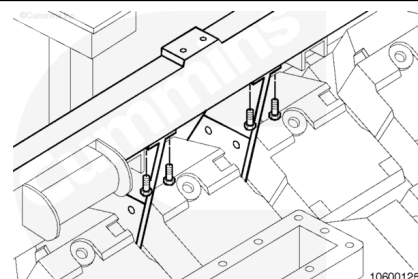
Remove the three capscrews from each of the two brackets connecting the low temperature aftercooling thermostat housing to the two-stage frame.



Remove the ten capscrews connecting the frame to each of the three center support brackets.

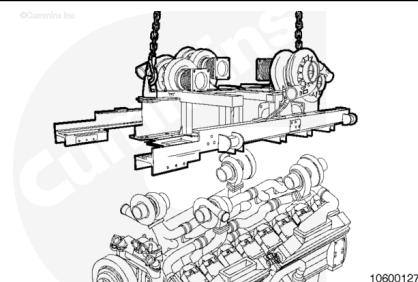


Remove the two capscrews connecting each of the outboard brackets to the two-stage frame.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



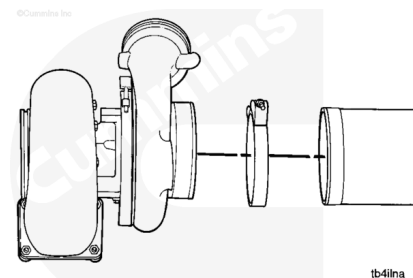
Connect the lifting straps to the two-stage frame lifting eyes.

Remove the two-stage frame from the engine.

with Electronically Actuated Injector

Disconnect the air intake piping between the air filter(s) and low pressure turbochargers.

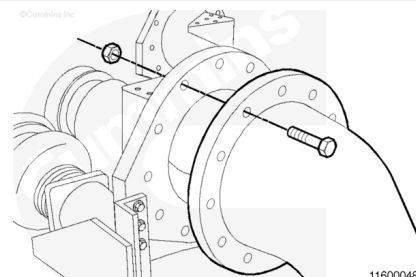
Disconnect the air intake piping at the inlet side of each turbocharger.



Remove the 12 exhaust outlet flange capscrews and nuts.

Disconnect the exhaust piping from the exhaust outlet flange.

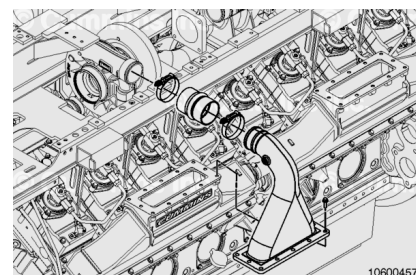
Repeat the above for the exhaust on the opposite side of the engine.



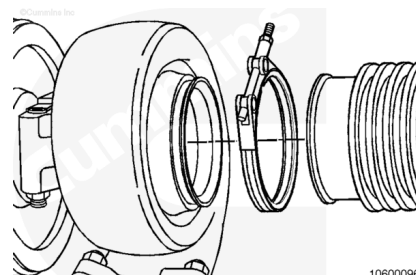
Remove the 12 capscrews connecting the air crossover to the aftercooler cover.

Remove the clamps from the hose connecting the air crossover to the high pressure turbocharger.

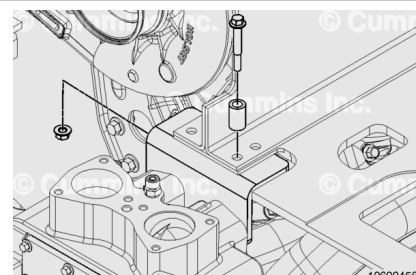
Remove the air crossover pipe.



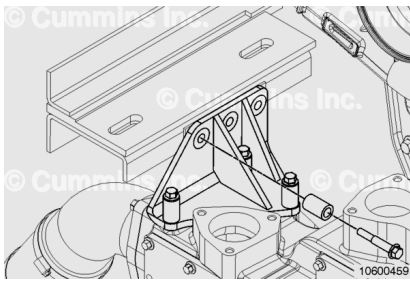
Disconnect the V-band clamp at the bellows on the turbine side of the four high-pressure turbochargers.



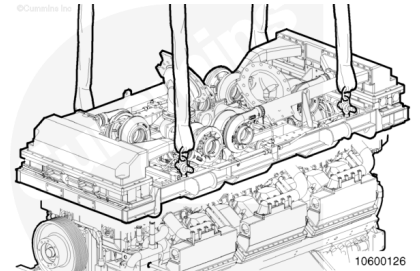
Remove the four capscrews from the bracket connecting the low temperature aftercooling housing to the two-stage frame.



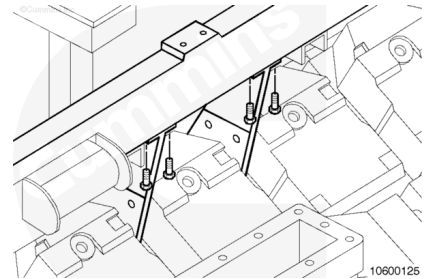
Remove the three capscrews from each of the two brackets connecting the low temperature aftercooling thermostat housing to the two-stage frame.



Remove the nine capscrews connecting the frame to each of the three center support brackets.

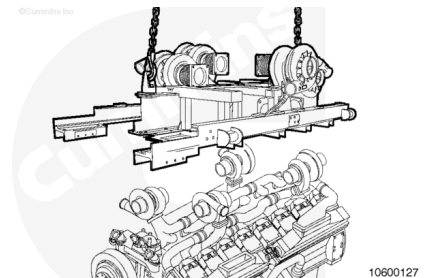


Remove the four capscrews connecting each of the outboard brackets to the two-stage frame.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



Connect the lifting straps to the two-stage frame lifting eyes.

Remove the two-stage frame from the engine.

Clean and Inspect for Reuse

Two Stage

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

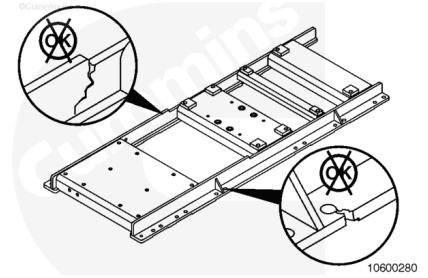
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the two-stage turbocharger frame and dry with compressed air.

Inspect the turbocharger frame for cracks and other damage.

Repair or replace the turbocharger frame if cracked or damaged.



Install

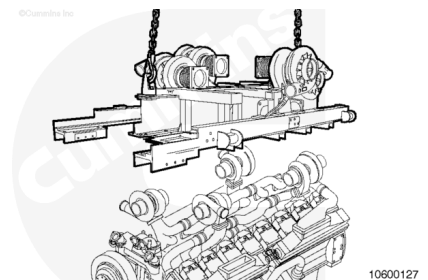
with Mechanically Actuated Injector

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Connect the lifting straps to the two-stage frame lifting eyes.

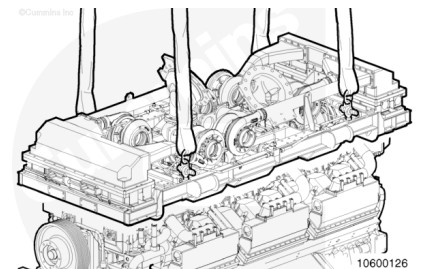
Position and align the two stage frame on the three center support brackets on the engine.



Do **not** tighten the two-stage frame assembly capscrews until all capscrews have been in the supports and brackets.

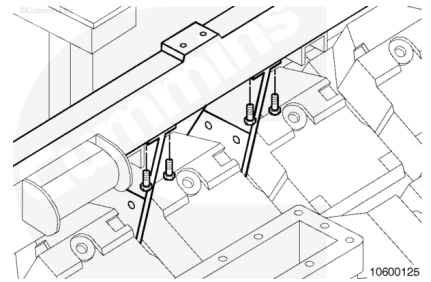
Align the two-stage frame assembly capscrew holes with the center support bracket capscrew holes.

Install the 10 capscrews connecting the two-stage frame to each of the center support brackets.



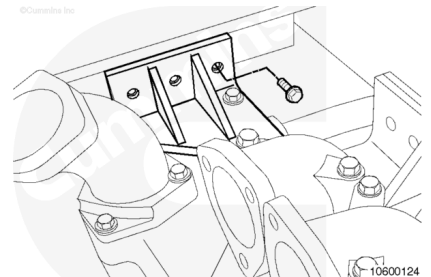
Align the two-stage frame assembly capscrew holes with the outboard support brackets.

Install the capscrews connecting the two-stage frame to each of the outboard support brackets.



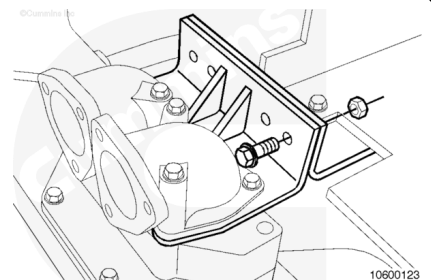
Align the two-stage frame assembly capscrew holes with the two low temperature thermostat housing bracket capscrew holes.

Install the capscrews connecting the two-stage frame to each of the two low temperature aftercooling thermostat housing brackets.

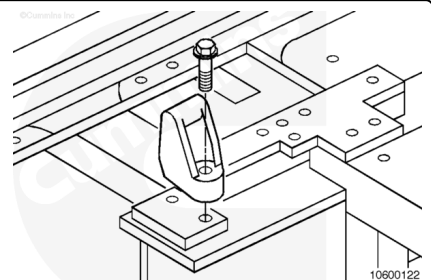


Tighten all of the two-stage frame assembly mounting capscrews.

Torque Value: 80 n•m [59 ft-lb]



Remove the lifting eyes from the two-stage frame.

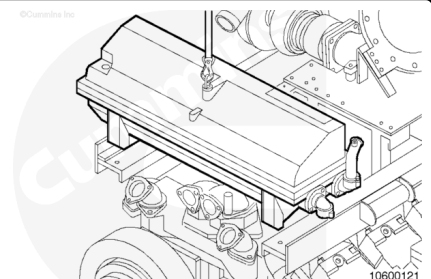


⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility injury, use a hoist or get assistance to remove this assembly.

Install a 12 mm [0.47 in] lifting device in the top of the intercoolers.

Install the Intercoolers.

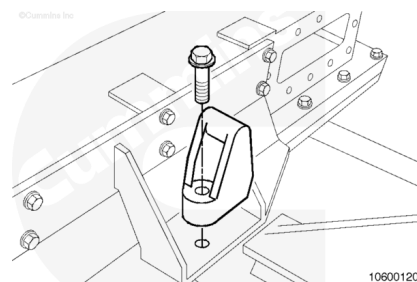


Remove the lifting eye from the intercoolers.

Install the lifting eye onto the two-stage frame assembly.

Tighten the lifting eye capscrew.

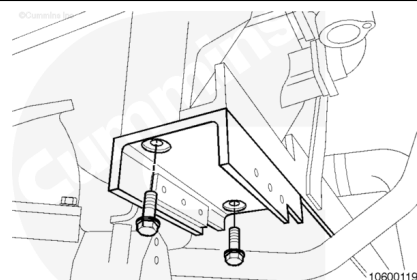
Torque Value: 195 n•m [144 ft-lb]



Install the intercooler mounting capscrews.

Tighten the capscrews.

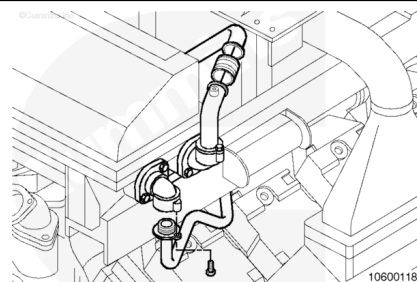
Torque Value: 280 n•m [207 ft-lb]



Install the coolant supply and return piping onto the intercoolers.

Tighten the capscrews.

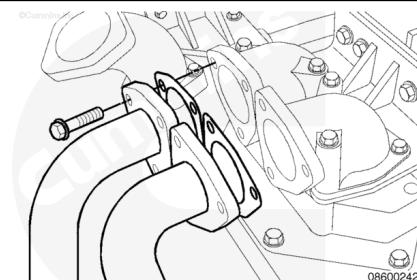
Torque Value: 45 n•m [33 ft-lb]



Install the low temperature aftercooling thermostat housing coolant supply lines.

Tighten the capscrews.

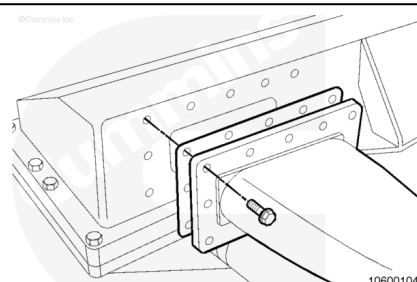
Torque Value: 54 n•m [40 ft-lb]



Install the four intercooler intake crossovers.

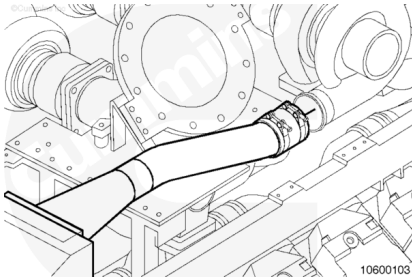
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

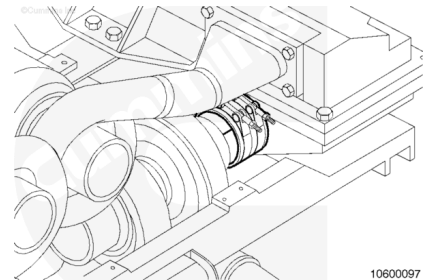


Install the hoses and clamps on the right-bank rear and left-bank front high-pressure turbochargers.

Torque Value: 9 n•m [80 in-lb]



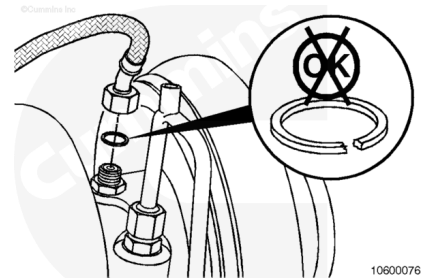
Install the intake hoses and clamps on each of the four high-pressure turbochargers.



Make sure the oil supply line o-ring is **not** damaged.

Install the oil supply lines on the four high-pressure turbochargers and tighten the fittings.

Torque Value: 54 n•m [40 ft-lb]

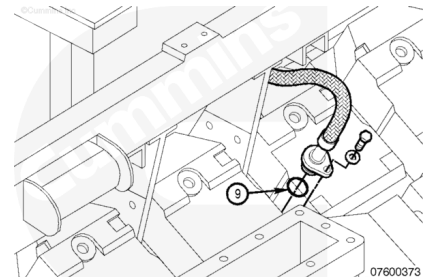


Install new o-ring seals on all of the turbocharger oil drain lines.

Install the oil drain lines into the cylinder block.

Tighten the oil drain line capscrews.

Torque Value: 45 n•m [33 ft-lb]

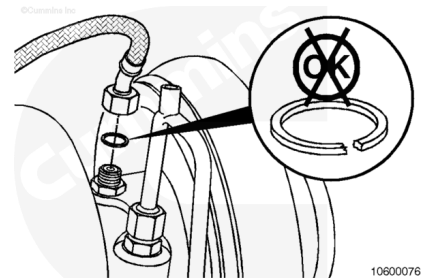


Make sure the oil supply line o-ring is **not** damaged.

Install the low-pressure turbocharger oil line to each low-pressure turbocharger.

Tighten the fittings.

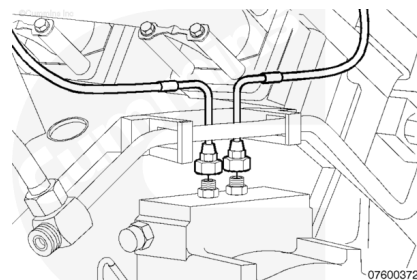
Torque Value: 54 n•m [40 ft-lb]



Connect the two oil supply lines at the flywheel housing.

Tighten the fittings.

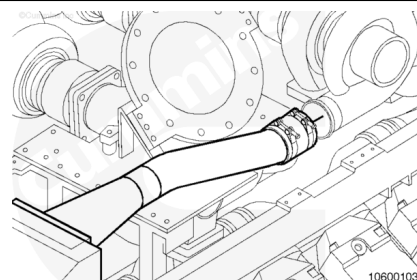
Torque Value: 54 n•m [40 ft-lb]



Install the air crossover that connect the left bank front and right bank rear high-pressure turbocharger to the aftercoolers.

Install the clamps on the hose connections at the air crossover to the high-pressure turbochargers.

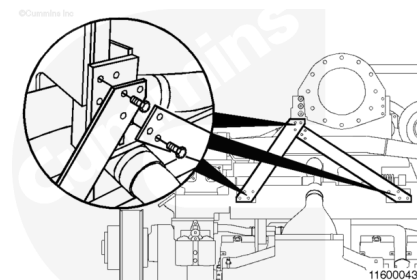
Torque Value: 9 n•m [80 in-lb]



Install the air crossover supports.

Tighten the capscrews.

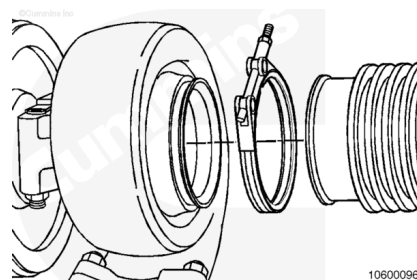
Torque Value: 45 n•m [33 ft-lb]



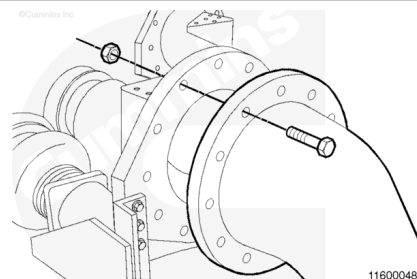
Install the V-band clamp and the exhaust bellows to the turbocharger exhaust outlet flanges.

Tighten the V-band clamp nut.

Torque Value: 9 n•m [80 in-lb]



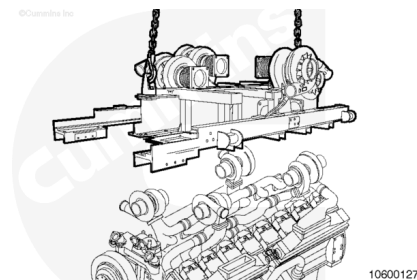
Connect the intake piping between the air filter(s) and the low-pressure turbochargers at the inlet side of the turbochargers.



with Electronically Actuated Injector

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



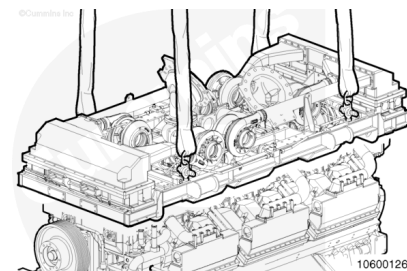
Connect the lifting straps to the two-stage frame lifting eyes.

Position and align the two stage frame on the three center support brackets on the engine.

Do **not** tighten the two-stage frame assembly capscrews until all capscrews have been in the supports and brackets.

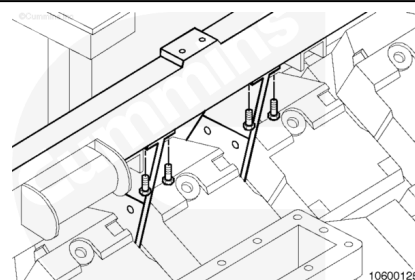
Align the two-stage frame assembly capscrew holes with the center support bracket capscrew holes.

Install the nine capscrews connecting the two-stage frame to each of the center support brackets.



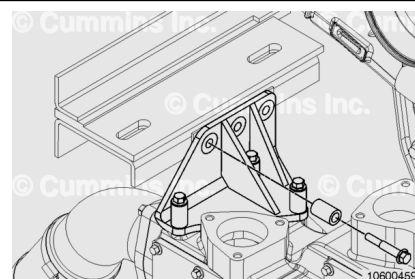
Align the two-stage frame assembly capscrew holes with the outboard support brackets.

Install the capscrews connecting the two-stage frame to each of the outboard support brackets.



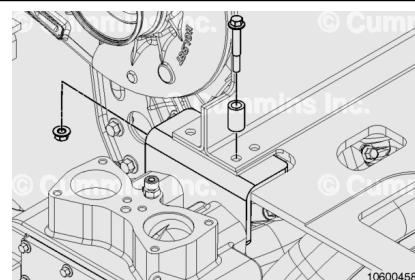
Align the two-stage frame assembly capscrew holes with the two low temperature thermostat housing bracket capscrew holes.

Install the capscrews connecting the two-stage frame to each of the two low temperature aftercooling thermostat housing brackets.

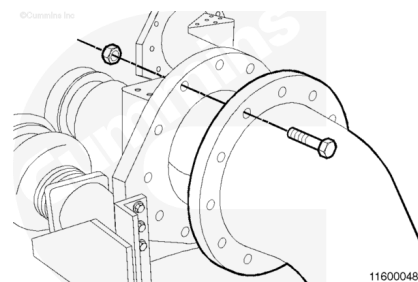


Tighten all of the two-stage frame assembly mounting capscrews.

Torque Value: 80 n•m [59 ft-lb]



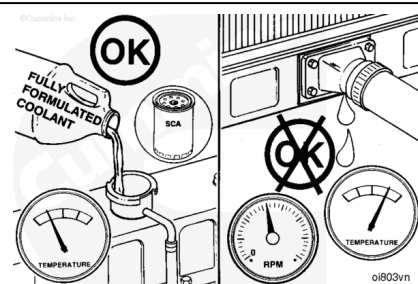
Connect the intake piping between the air filter(s) and the low-pressure turbochargers at the inlet side of the turbochargers.



Finishing Steps

with Mechanically Actuated Injector

- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).



with Electronically Actuated Injector

- Install the intercooler assembly. Refer to Procedure 010-083 (</qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html>).
- Install the turbocharger cooling hoses. Refer to Procedure 010-041 (</qs3/pubsys2/xml/en/procedures/56/56-010-041.html>).
- Install the turbocharger oil return lines. Refer to Procedure 010-045 (</qs3/pubsys2/xml/en/procedures/56/56-010-045-tr.html>).
- Install the turbocharger oil supply lines. Refer to Procedure 010-046 (</qs3/pubsys2/xml/en/procedures/56/56-010-046-tr.html>).
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

